Simple Linear Regression Model

Regression analysis is a statistical technique that is useful in exploring the relationships between two or more variables.

In the simple linear regression model the dependent variable, or response, is related to one independent or regressor variable as

$$Y = \beta_0 + \beta_1 x + \epsilon$$

where ϵ is the random error term. The parameters β_0 & β_1 are called regression coefficients.

Scatter → X - independent or regressor variable

Y → response or dependent variable.

estimate the β_0, β_1 by maximum likelihood, properties of least squares, method of moments etc.

The method of max. likelihood needs further knowledge of the distn. of y whereas the method of moments & the principle of least squares do not need any knowledge about the distn of y.

Regression means "to move in backward direction".

Model exists in nature but is unknown to experiments. When some values to the explanatory variables are provided, then the values for the o/p or study variable are generated depending on the form of the fn. f & the nature of phenomenon. So ideally, the preexisting model give rise to the data. Our objective is to det. the final form of this model. Now we move in the backward direction. We propose to first collect the data on study & explanatory variables. Then we

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The variance of Y given X is

$$V(Y/x) = V(\beta_0 + \beta_1 x + \epsilon) = V(\beta_0 + \beta_1 x) + V(\epsilon) = 0 + \sigma^2 = \sigma^2$$

The true regression model is a line of mean values:

$$\mu_{Y/x} = \beta_0 + \beta_1 x$$

Where β_1 can be interpreted as the change in mean of Y for a unit change in x.

Also the variability of Y at a particular value of x is determined by error variance, σ^2.

$\Rightarrow$ there is distn of Y values at each x & that variance of this distn is same at each x.

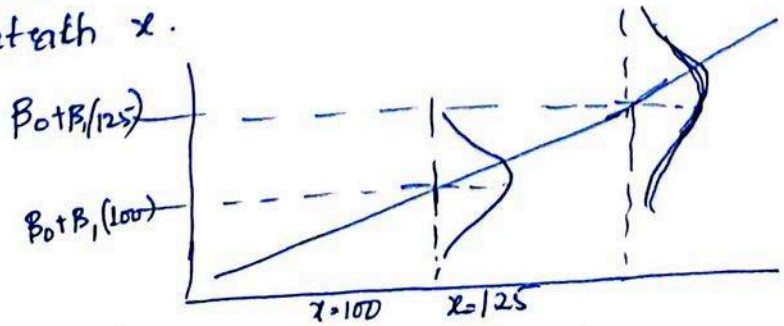
$$Y = \beta_0 + \beta_1 x + \epsilon$$

This model is called simple linear regression model.

True regression model

$\mu_{Y/x} = \beta_0 + \beta_1 x$ is a line of mean values, i.e ht of the regression line at any value of x is just the expected value of Y for that x. The slope β_1, can be interpreted as the change in the mean of Y for a unit change in x.

Furthermore, the variability of Y at a particular value of x is determined by the error variance σ^2. This $\Rightarrow$ that there is a distn of Y values at each x and that the variance of this distn is the same at each x.



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$E(Y|x) = \beta_0 + \beta_1 x$.

where the intercept β_0 & the slope β_1 are unknown regression coefficients. We assume that each obsvn Y can be described by the model

$$Y = \beta_0 + \beta_1 x + \epsilon$$

where ϵ is a random error with mean zero & (unknown) variance σ^2.

Karl

The German scientist, Gauss (1777-1855) proposed estimating the parameters β_0 & β_1 to min the sum of squares of vertical devn & is called as method of least squares.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \qquad i = 1, 2, \ldots, n. \qquad \text{— (1)}$$

& sum of squares of devns of obsvns from true regression line is

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 \qquad \text{— (2)}$$

The least sq estimators of β_0 & β_1 say $\hat{\beta}_0$ & $\hat{\beta}_1$ must satisfy

$$\left.\frac{\partial L}{\partial \beta_0}\right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0. \qquad \text{— (3)}$$

$$\left.\frac{\partial L}{\partial \beta_1}\right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0. \qquad \text{— (4)}$$

Simplifying these 2 eqns yield.

$$\left.\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i &= \sum_{i=1}^{n} y_i \\ \hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} x_i y_i \end{aligned}\right\} \qquad \text{— 5}$$

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The least sq. estimates of the intercept & slope are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} \quad \text{---- (6)}$$

where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ & $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$.

The fitted or estimated regression line is:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Note for each pair of obsrns satisfies the relnship.

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x + e_i \quad i = 1, 2, \ldots n$$

where $e_i = y_i - \hat{y}_i$ is called the residual. It describes the error in the fit of the model to the ith obsvn y_i.

Notationally, it is occasionally convenient to give sp. symbols to the numerator & denominator of eqn (6).

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} \quad \text{---- (7)}$$

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})^2 * y_i = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i\right)}{n}$$

Estimating σ^2.

There is actually another unknown parameter in regression model σ^2 (the variance of error term ϵ). The residual $e_i = y_i - \hat{y}_i$ are used to estimate σ^2. The sum of squares of the residuals, often called error sum of squares is,

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We can show that expected value of error sum of squares is

$$E(SS_E) = (n-2)\sigma^2.$$

∴ an unbiased estimator of σ^2 is.

$$\sigma^2 = \frac{SSE}{n-2} \quad \text{——} \quad (9).$$

$$SS_E = SS_T - \hat{\beta}_1 S_{xy}$$

where $SST = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2$ is total sum of squares of response variable y.

Regression methods were used to analyse the data from a study investigating the relationship b/w roadway surface temp (x) & pavement deflection (y). Summary quantities were

$$n = 20 \quad \Sigma y_i = 12.75 \quad \Sigma y_i^2 = 8.86 \quad \Sigma x_i = 1478$$

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$= 143,215.8 - \frac{(1478)^2}{20} = 33991.6.$

$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} = 1083.67 - \frac{12.75 * 1478}{20}$

$= 141.445.$

$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{141.445}{33991.6} = 0.00416.$

$\hat{\beta}_0 = \bar{y} - \beta_1 \bar{x} = \frac{12.75}{20} - (0.00416) * \frac{1478}{20}$

$= 0.32999.$

$\hat{y} = 0.32999 + 0.00416x.$

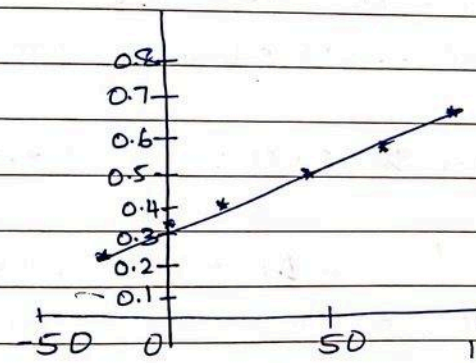
$\sigma^2 = MS_E = \frac{SSE}{n-2} = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n-2} = \frac{\left(\sum_{i=1}^{n} y_i^2 - n\bar{y}^2\right) - \hat{\beta}_1 S_{xy}}{n-2}$

$= \frac{\left(8.86 - 20 * \left(\frac{12.75}{20}\right)^2 - 0.00416 * 141.445\right)}{18}$

$= \frac{0.143}{18}$

$= 7.91 * 10^{-3}$

$= 0.00796$



b. $\hat{y} = 0.32999 + 0.00416 * (85) = 0.6836.$

c. $\hat{y} = 0.32999 + 0.00416 * (90) = 0.7044.$

d. A change of 0.00416 in mean pavement deflection would be expected for 1°F change in surface temp. ∵ 0.00416 is the slope & slope is avg increase/decrease of y per unit of x.

2. Suppose the temp is measured in °C rather than °F. Write the new regression model that results.

b) What change in expected pavement deflection is associated

$\hat{y} = 0.4631476 + 0.0074902x$

$\hat{\beta}_1$ = Slope = 0.0074902x.

An article presented a data on compressive strength x & intrinsic permeability y of various concrete mixes & cure Summary qtyties are

$n = 14 \quad \Sigma y_i = 572 \quad \Sigma y_i^2 = 23530 \quad \Sigma x_i = 43,$

$\Sigma x_i^2 = 157.42$ & $\Sigma x_i y_i = 1697.80$. Assume that the & variables are related according to the simple LRM.

Calculate the least squares estimates of the slope & the intercept

Use the eqn of the fitted line to predict what permeability would be observed when the compresive strength is 43

Give a point estimate of the mean permeability when compressive strength is $x = 3.7$.

Suppose that the observed value of permeability is $y = 46.1$ at $x = 3.7$. Calculate the value of corresponding residual.

$y_i = \beta_0 + \beta_1 x + \epsilon_i$.

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$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{572}{14} - (-2.3298017)\left(\frac{43}{14}\right)$$

$$= 48.013.$$

b) $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$

$\hat{y} = 48.013 - 2.3298017 * 4.3 = 37.99.$

c. $\hat{y} = 48.013 - 2.3298017\,(3.7) = 39.39.$

d. $e = y - \hat{y} = 46.1 - 39.39 = 6.71.$

11.8 (3). Data on green liqor Na_2S conc (in gms/lt) & paper m/c prodn (in tons/day). The data are as follows.

y	40	42	49	46	44	48	46	43	53	52	54	57	58
x	825	830	890	895	890	910	915	960	990	1010	1012	1030	1050

a. Fit a simple LRM with y = green liquors Na_2S conc & x = prodn. Find an estimate of σ^2. Draw a scatter diagram of the data & resulting least squares fitted model.

b) Find the fitted value of y corresponding to x = 910 & the associated residual.

c). Find the mean green liquor Na_2S conc when prodn rate is 950 tons/day.

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